

# JEE MAIN REVISION MASTER: HYDROCARBONS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 166 Analyzed PYQs

## 1. High-Yield Concept Priority Matrix

| Priority Tier | Concept Name                                              | Frequency | Category              |
|---------------|-----------------------------------------------------------|-----------|-----------------------|
| Very High     | Hydrocarbons — Applied Cases, Reactions & Traps           | 129 PYQs  | Applied Topics        |
| Very High     | Hydrocarbons — Standard Calculations & Problem Archetypes | 36 PYQs   | Quantitative Analysis |
| Very High     | Hydrocarbons — Core Principles & Governing Laws           | 15 PYQs   | Core Principle        |

## 2. Master Formula Reference Card

| Concept / Topic                                           | Mathematical Formulation & Governing Equations                                                                                          |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Hydrocarbons — Applied Cases, Reactions & Traps           | $C_xH_y(g) + \frac{4x+y}{4}O_2(g), \quad \pi=4, \quad C_{10}H_{14}$                                                                     |
| Hydrocarbons — Standard Calculations & Problem Archetypes | $(Ph-CH=CH-CH_3) \rightarrow cis(Ph-CH=CH-CH_3), \quad H_2\overset{\oplus}{N}O_3 \rightarrow H_2O + \overset{\oplus}{N}O_2, \quad S_N1$ |
| Hydrocarbons — Core Principles & Governing Laws           | $\frac{y}{2}, \quad \frac{y}{4}, \quad _9$                                                                                              |

## 3. Deep-Dive Concepts & Linked PYQ Evidence

### 3.1 Hydrocarbons — Applied Cases, Reactions & Traps (Very High Yield — 129 PYQs)

**Core Principle:** Special cases, experimental observations, and conceptual traps designed to test deep understanding of Hydrocarbons.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

**Representative Source PYQs:**

- [2026 Evening Shift] Q#1 (MCQ): "The major product of which of the following reaction is not obtained by rearrangement reaction?..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "The dibromo compound [P] (molecular formula:  $C_{(9)}H_{(10)}Br_{(2)}$ ) when heated with excess sodamide followed by treatment with dilute HCl ..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (Numerical): "Consider the following reaction of benzene. In compound (Q), the percentage of oxygen is \_\_\_\_ %. (Nearest integer)..." [Ans: 10]

### 3.2 Hydrocarbons — Standard Calculations & Problem Archetypes (Very High Yield — 36 PYQs)

**Core Principle:** Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Hydrocarbons.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

**Representative Source PYQs:**

- [2026 Evening Shift] Q#1 (MCQ): "Identify the correct IUPAC name of hydrocarbon (x) containing three primary carbon atoms and with molar mass  $72\text{g mol}^{-1}$  ...." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Arrange the following alkenes in decreasing order of stability. Choose the correct answer from the options given below:..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (Numerical): "Consider the following reaction of benzene. In compound (Q), the percentage of oxygen is \_\_\_\_ %. (Nearest integer)..." [Ans: 10]

### 3.3 Hydrocarbons — Core Principles & Governing Laws (Very High Yield — 15 PYQs)

**Core Principle:** Fundamental definitions, core postulates, and governing physical/chemical relations in Hydrocarbons.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Hydrocarbons.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Hydrocarbons PYQs.

**Representative Source PYQs:**

- [2026 Morning Shift] Q#3 (MCQ): "Given below are two statements : Statement I : Methane can be prepared by decarboxylation of sodium ethanoate, Kolbe's electrolysis of sodiu..." [Ans: D]
- [2026 Morning Shift] Q#3 (MCQ): "An alkene(X) on ozonolysis followed by reduction gives following products. The alkene(X) is :..." [Ans: N/A]
- [2026 Evening Shift] Q#3 (MCQ): "The final product[B] is :..." [Ans: N/A]

## 4. Benchmark Illustrative PYQs with Master Solutions

**Benchmark Problem 1: JEE Main 2026 (Online) 23rd January Evening Shift (Q#1)**

Consider the following reaction of benzene. In compound (Q), the percentage of oxygen is \_\_\_\_ %. (Nearest integer)

**Correct Answer: 10**

**Solution Analysis:** Molecular mass of compound Q is 157. In Q, oxygen contributes mass 16 (for one oxygen atom). So, percentage of oxygen in Q is % of oxygen in product 'Q' is  $= (16 / 157) \times 100 = 10.19\%$  Nearest integer = 10 %.

**Benchmark Problem 2: JEE Main 2026 (Online) 6th April Morning Shift (Q#3)**

Given below are two statements : Statement I : Methane can be prepared by decarboxylation of sodium ethanoate, Kolbe's electrolysis of sodium acetate and reaction of  $\text{CH}_3\text{MgBr}$  with water. Statement II : Methane cannot be prepared from unsaturated hydrocarbons and by Wurtz reaction. In the light of the above statements, choose the correct answer from the options given below

- (A) Both Statement I and Statement II are true
- (B) Both Statement I and Statement II are false
- (C) Statement I is true but Statement II is false
- (D) Statement I is false but Statement II is true **[Correct]**

**Correct Answer: D**

**Solution Analysis:** Analysis of Statement I Statement I says: "Methane can be prepared by decarboxylation of sodium ethanoate, Kolbe's electrolysis of sodium acetate and reaction of  $\text{CH}_3\text{MgBr}$  with water." Decarboxylation of sodium ethanoate: This reaction involves heating a sodium salt of a carboxylic acid with soda-lime (a mixture of NaOH and CaO). The general reaction is the removal of a carboxyl group. For sodium ethanoate ( $\text{CH}_3\text{COONa}$ ), the reaction is:  $\text{CH}_3\text{COONa} + \text{NaOH} \rightarrow \text{CH}_4 + \text{Na}_2\text{CO}_3$  As you can see, methane ( $\text{CH}_4$ ) is indeed the product. So, this

**Benchmark Problem 3: JEE Main 2025 (Online) 29th January Morning Shift (Q#4)**

The sum of sigma ( $\sigma$ ) and pi ( $\pi$ ) bonds in Hex-1,3-dien-5-yne is \_\_\_\_\_.

**Correct Answer: 15**

**Solution Analysis:** The compound given is hex-1,3-dien-5-yne structure is A double bond has one  $\pi$  bond and one sigma bond. A triple bond has one sigma bond and two  $\pi$  bonds. For the one triple bond, number of  $\pi$  bonds = 2 For one double bond, number of  $\pi$  bond = 1 For two double bonds, number of  $\pi$  bonds = 2 So, total number of  $\pi$  bonds =  $2 + 2 = 4$  For sigma bonds, count all carbon - carbon bonds and carbon - hydrogen bonds except the  $\pi$  bonds in double bonds and triple bond. The total number of sigma bonds = 11 So, total number of bonds:  $\sigma = 11$   $\pi = 4$  Sum of  $\sigma$  and  $\pi$  bonds =